

Solution Architecture Document

ARC — Agentic Receivables Recovery and Section 138 Legal Action Pipeline

Attribute	Value
Solution	ARC
Primary Reference	Problem_Statement_1_Receivables_Recovery_Section138 (Organization-approved)
Stack	C# · .NET 9 · Microsoft Agent Framework 1.0 · Azure
Version	1.0
Date	28-Aug-2026
Status	Architecture — Pre-Implementation

Architecture principle: AI reasoning, orchestration and explanation sit above governed tools. Money, dates, eligibility, authorization and persistence are always deterministic — rules engine, metric contracts, SQL/graph queries, and human gates. The LLM never computes legal amounts, statutory dates, or eligibility verdicts.

1. Architecture Overview

ARC is a **layered, durable, human-gated agentic workflow system** for PaintCo receivables recovery. It implements two Microsoft Agent Framework (MAF) workflows sharing upstream reconciliation (A1), eight specialised executors (A1–A8), a versioned rules engine (R1–R6), governed tools, and four mandatory human approval gates.

Layer	Responsibility	Technologies
Presentation / Integration	Gate UI, dashboards, NLQ, field app hooks	ARC.Api (Container Apps), Entra ID
Workflow Orchestration	Multi-day MAF graphs, RequestPort suspensions, fan-out/fan-in	ARC.Host.Functions, MAF Durable
Agents / Executors	A1–A8 — bounded reasoning, tool selection, drafting	ARC.Agents
Governed Tools	Typed tool surface + MCP	ARC.Tools
Business Services	Rules engine, limitation clock, metric contracts, outbound gate	ARC.Domain + ARC.Data
Knowledge / Retrieval	Graph, hybrid search, document extraction	ARC.Knowledge
Data	Master data, workflow state, documents	Azure SQL, Cosmos DB, Blob
External Systems	ERP ledger, field app, advocate/court (manual beyond filing readiness)	SAP, Portal, Field App

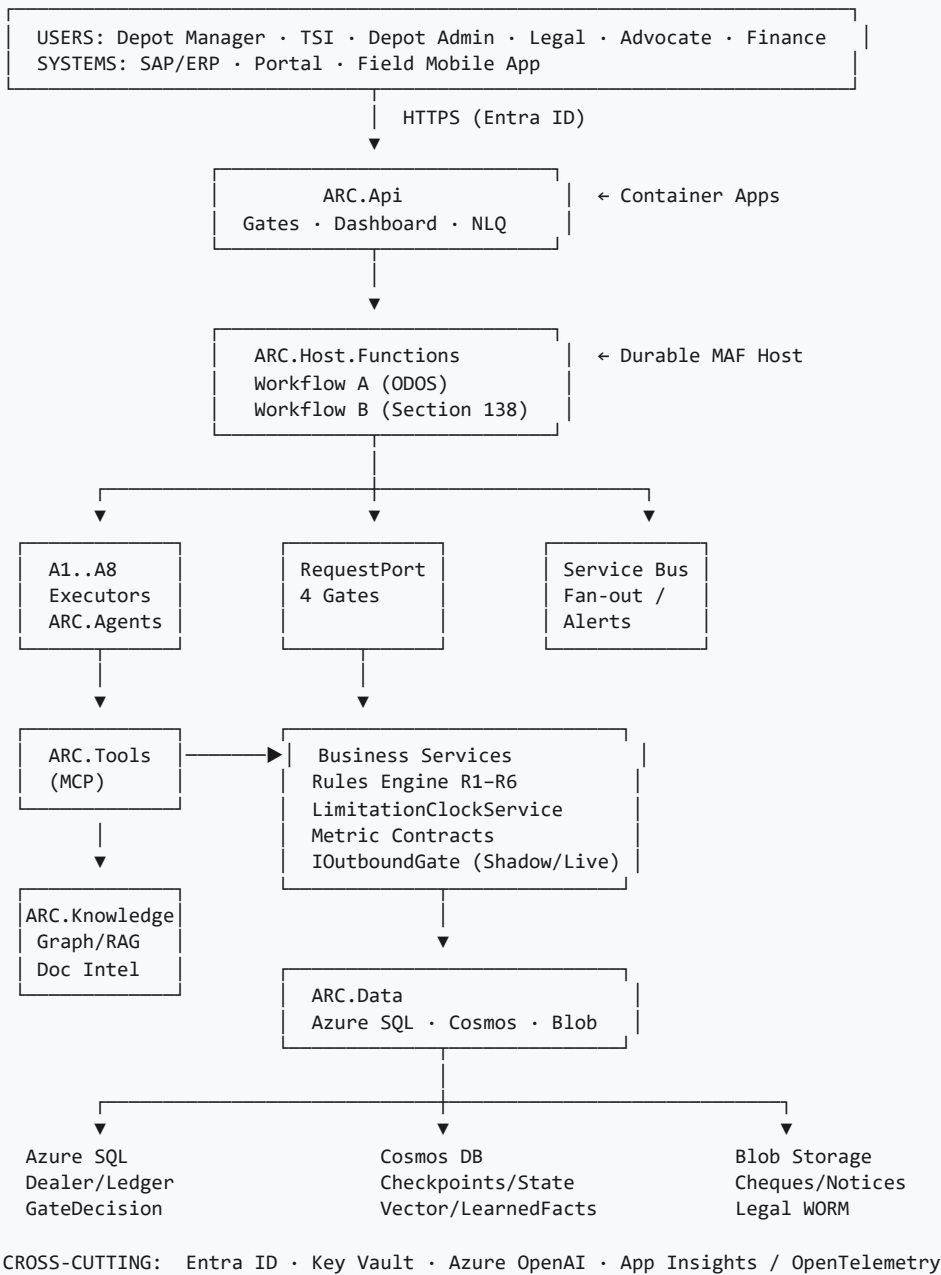
Layered flow:

User / Business Systems → ARC API → MAF Workflow Orchestration → Agents/Executors → Governed Tools → Business Services/Rules → Data & External Systems

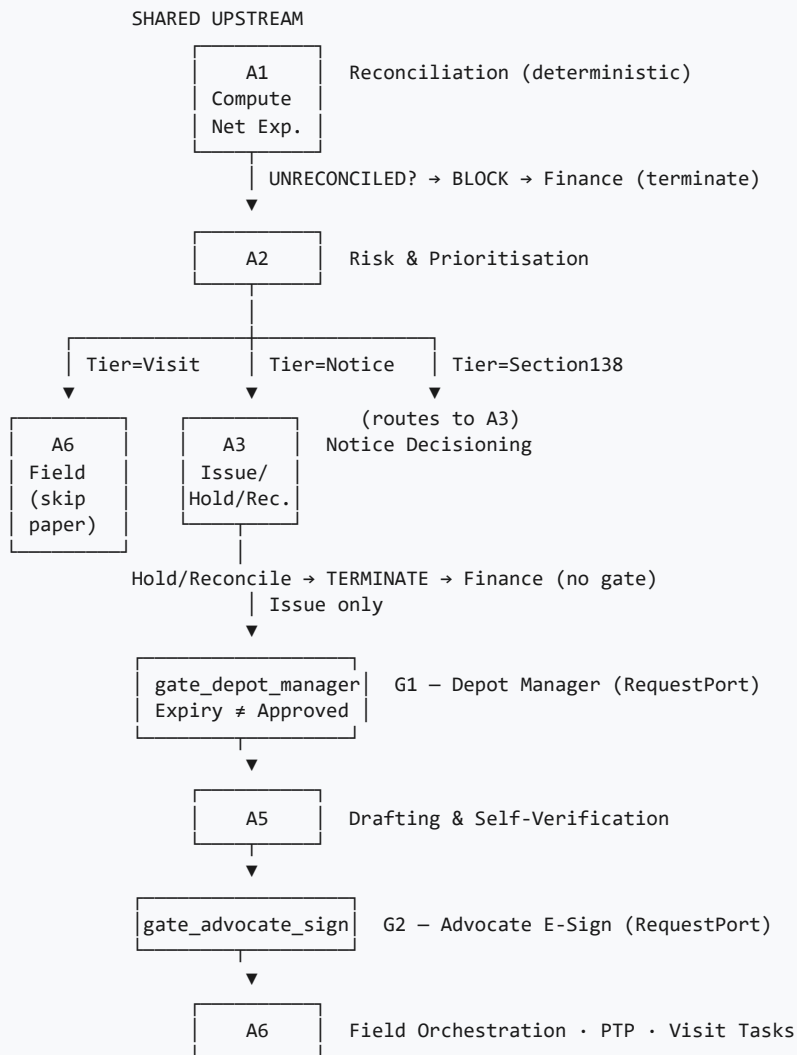
1.1 Mandated Technical Stack (from Source)

Concern	Service / Package
Runtime	C# · .NET 9
Orchestration	Microsoft Agent Framework 1.0 — WorkflowBuilder, RequestPort, durable checkpoints
Models	Azure OpenAI (Foundry) — reasoning for A3/A4 explanation; cheaper model for extraction/summary
Master data	Azure SQL
Workflow state / vector	Azure Cosmos DB (checkpoints, cycle state, documents, learned facts, audit events)
Documents	Azure Blob (WORM legal artefacts, 8-year retention)
Document AI	Azure AI Document Intelligence
Voice PTP	Azure AI Speech
Compute	Azure Functions (durable host) + Azure Container Apps (API)
Messaging	Azure Service Bus (cycle fan-out, alerts)
Identity / Secrets	Microsoft Entra ID, managed identity, Azure Key Vault
Observability	Application Insights + OpenTelemetry
Resilience	Polly (retry/circuit breaker)
Tool protocol	MCP server exposing same contracts as AIFunction tools

2. Final Solution Architecture Diagram

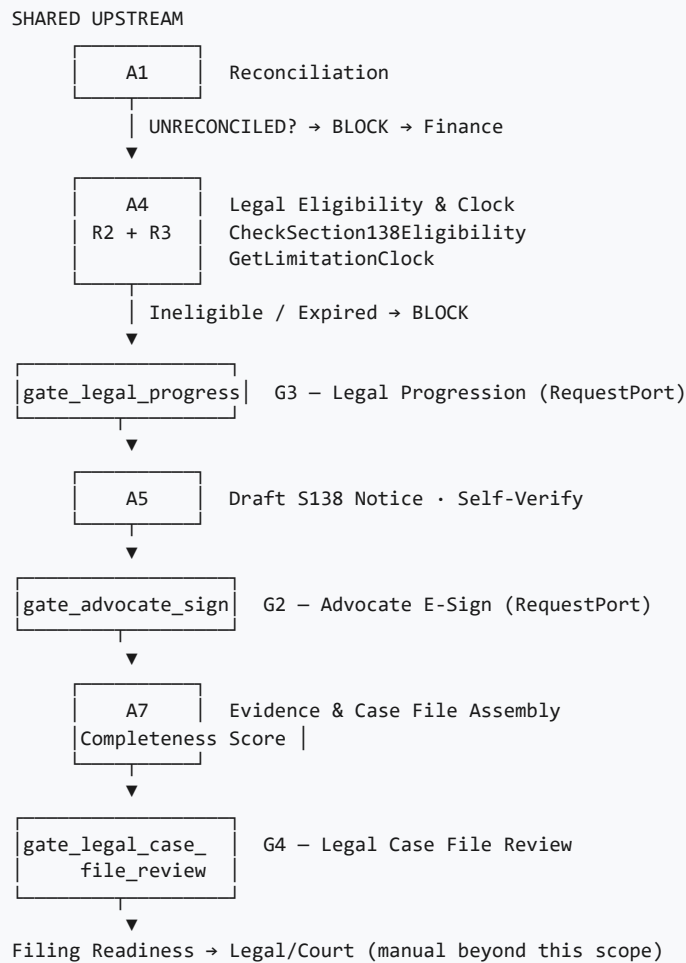


3. ODOS Workflow Diagram (Workflow A)



Terminal outputs: WithoutOutputFrom(A3 terminated, A6 completed)
Idempotency key: (cycle_id, dealer_urn, node)
Duration: 15+ calendar days (1st → 7th approval → ongoing)

4. Section 138 Workflow Diagram (Workflow B)



5. A1–A8 Agent Architecture

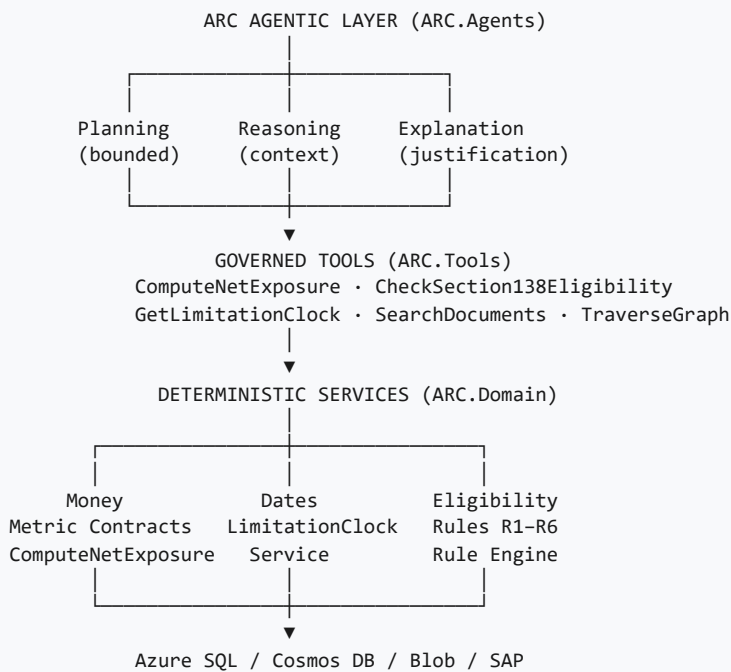
Agent	Purpose	Input	Output	Deterministic / AI	Gate
A1 Reconciliation	Net recoverable exposure with lineage	DealerUrn, AsOf, SAP ledger	ExposureBreakdown + LinItemRef trail	100% Deterministic	None
A2 Risk & Prioritisation	Rank dealers; assign recovery tier	Exposure, graph features, TSI remarks	tier ∈ {Visit, Notice, Section138}	Hybrid — scoring deterministic; LLM reads qualitative remarks	None
A3 Notice Decisioning	Evidence-backed notice verdict	RecoveryState + Exposure	Issue / Hold / Reconcile + citations	Rules decide; LLM phrases justification only	G1 Depot Manager (Issue only)
A4 Legal Eligibility & Clock	S138 eligibility + statutory dates	ChequeReturnMemo, DemandNotice	Eligible/blocked + LimitationClock + alerts	Rules engine decides; LLM explains only	G3 Legal

A5 Drafting & Verification	Fill templates; self-verify fields	Graph facts, templates, return memo	Draft PDF for e-signature	Template fill + verification deterministic	G2 Advocate
A6 Field Orchestration	Visit plans; voice → PTP; chase	Risk tier, location, voice input	Visit tasks, PTP records, escalations	Hybrid — ASR+LLM for voice; TSI confirms low confidence	TSI confirm (not RequestPort)
A7 Evidence & Case File	Assemble court-ready bundle	All evidence artefacts	CaseFile + completeness score	Mostly deterministic assembly/scoring	G4 Legal
A8 Supervisory Insight	Exceptions; lever effectiveness; NLQ	Semantic layer, cycle data	Exception queue, analytics	LLM for NLQ; analytics deterministic	None

Not eight unrestricted autonomous agents. Each executor has a bounded contract. A1, rules evaluation, clock computation, and A5 self-verification are deterministic. LLM is used for explanation, summarisation, bounded planning, and voice parsing — never for money, dates, or eligibility verdicts.

Cheque selection (deterministic): Newest bounced cheque with return memo; else newest on file. Never delegated to the model.

6. AI vs Deterministic Responsibility Diagram



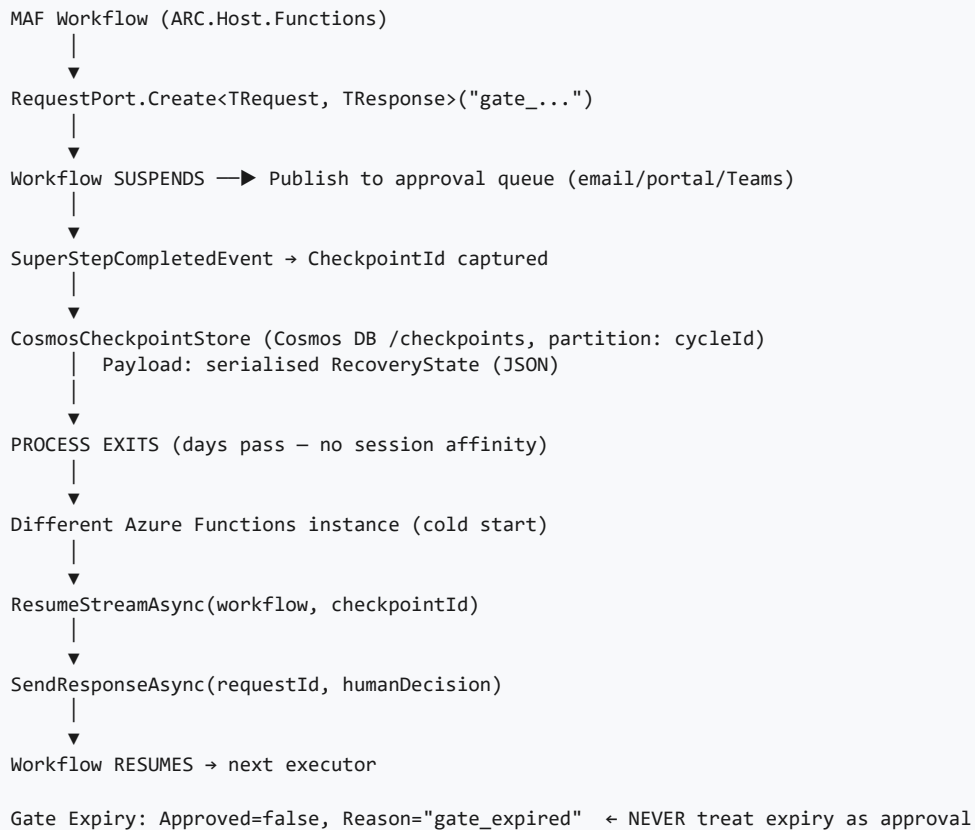
Document Intelligence: schema-validated extraction (confidence-gated) – NOT LLM math
Azure AI Speech + LLM: vernacular voice → structured PTP – TSI confirms before commit

LLM DOES NOT:

- Calculate legal amounts
- Calculate statutory dates
- Decide Section 138 eligibility
- Approve notices or legal action
- Bypass authorization
- Directly update databases
- Call SubmitNotice (tool does not exist)

Concern	Handler
Money, dates, eligibility, clocks	Rules engine + SQL/graph — never LLM
Cheque/memo/POD extraction	Document Intelligence + confidence gates
Summarisation, justification, narratives	LLM (from deterministic outputs only)
Planning, routing, tool selection	LLM supervisor, bounded by tool contract
Voice → PTP	Azure AI Speech + LLM → TSI confirmation

7. Human-in-the-Loop / Checkpoint Architecture



Gate ID	Approver	Trigger	Checkpoint
gate_depot_manager	Depot Manager	A3 verdict = Issue	Cosmos DB
gate_advocate_signature	Advocate	A5 self-verification pass	Cosmos DB
gate_legal_progression	Legal Team	A4 eligibility pass (Workflow B)	Cosmos DB
gate_legal_case_file_review	Legal Team	A7 assembly complete	Cosmos DB

R4: Agent that recommends cannot approve. **Audit:** GateDecision table (SQL) + Cosmos auditEvents — immutable, 8-year archive.

In-memory checkpointing is NOT sufficient. Cycle spans 15+ days; must survive process restart on different instance.

8. Data Architecture

8.1 Azure SQL — Master / Transactional (Authoritative)

Entity / Table	Responsibility
Dealer	Resolved identity (DealerUrn), SAP/Portal/App IDs, moratorium (R5), depot/region/TSI
LedgerPosition	Invoices, credit notes, payments, rebates, returns; ageing on DueDate; SourceRowKey (R6)
SecurityCheque	Cheque particulars, status, blob path, extraction confidence
ChequeReturnMemo	Return reason raw/code, memo dates (clock anchor), blob path
DemandNotice / Section138Notice	Notice lifecycle, amounts (from metric contract only)
GateDecision	Immutable human gate audit — actor, role, decision, override, correlation ID
FieldVisit / PromiseToPay / Dispute	Recovery tracking

Pattern: EF Core / Dapper, **read-only by default** for agent queries. Writes via business services only.

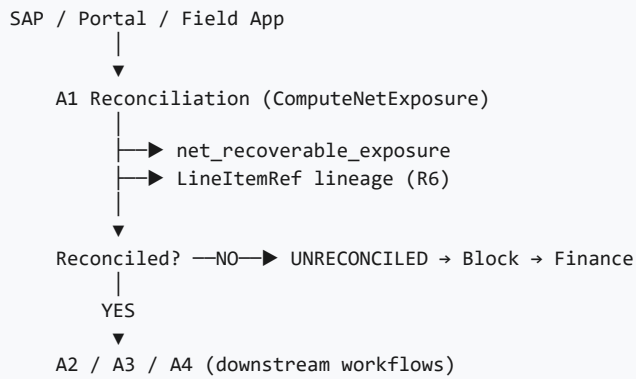
8.2 Azure Cosmos DB — Workflow, Semantic, Vector

Container	Partition Key	Contents	TTL
checkpoints	/cycleId	MAF workflow checkpoints (RecoveryState JSON)	90 days
cycleState	/cycleId	Per-dealer run state, terminal outcomes	None
documents	/documentType	Chunked policies, notices, agreements + embeddings (3072-dim, cosine, diskANN)	None
learnedFacts	/scope	A-Box enrichment with confidence and expiry	365 days
auditEvents	/cycleId	Append-only agent/tool event log	8 years (archive)

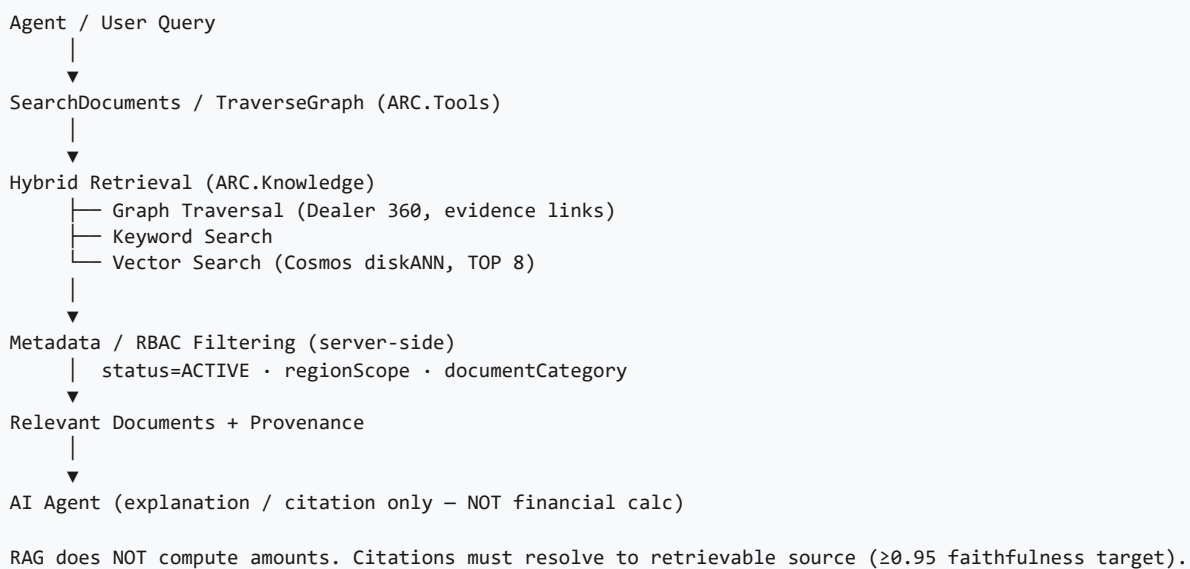
8.3 Azure Blob Storage — Documents

Content	Notes
Cheque images, return memos, PODs	Document Intelligence input
Demand notices, S138 notices	A5 output; WORM container
Dealer agreements, service proof	Evidence for A7
Legal documents / evidence bundles	WORM/immutable, 8-year retention

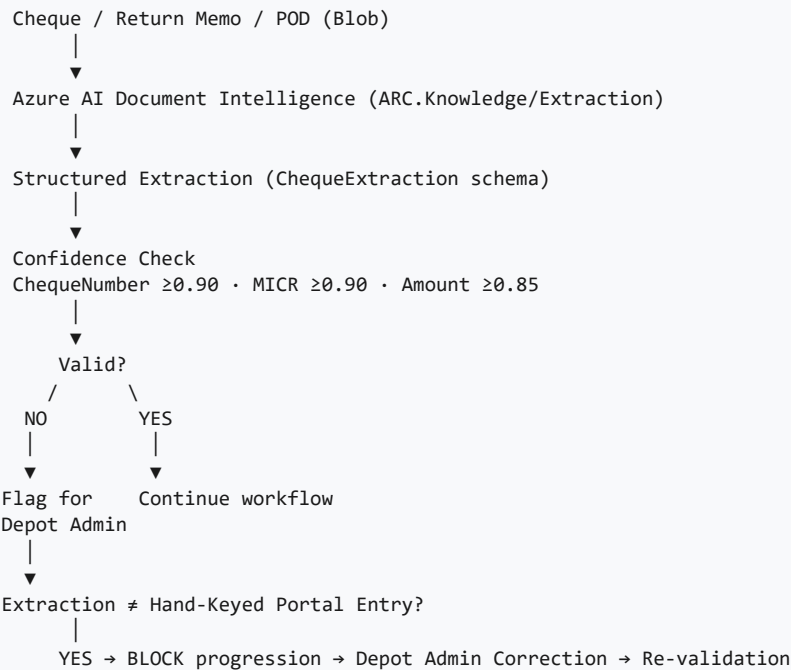
8.4 Shared Upstream — A1 Data Flow



9. Knowledge / RAG Architecture



10. Document Intelligence Architecture



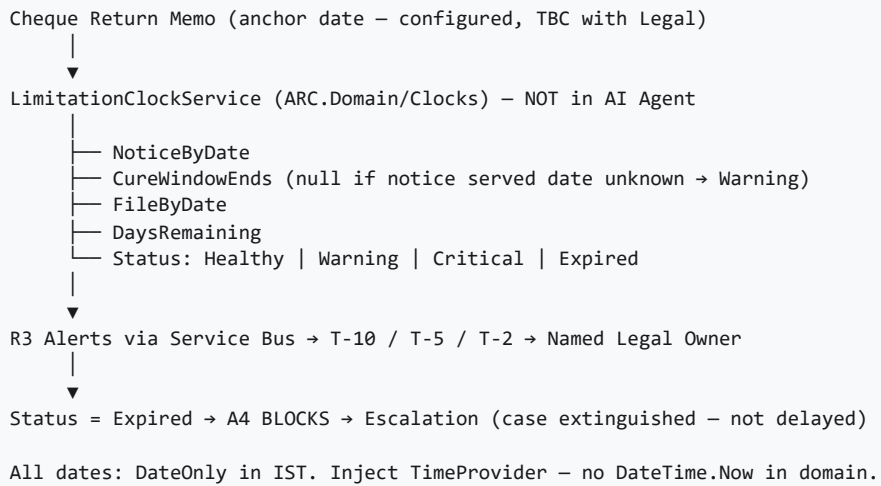
11. Rules Engine Architecture

Separate versioned component in **ARC.Domain/Rules**. AI agents invoke rules; they never replace them.

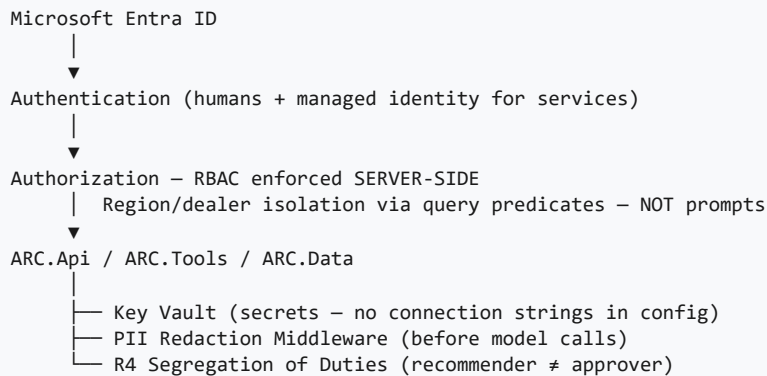
Rule	Description	Owner
R1a	Hold if unreconciled	Finance / Rules Engine
R1b	Hold if unapplied_credits / claim_amount > 0.40 (configurable)	Finance
R1c	Hold if Dispute UNDER_REVIEW or active PTP within grace	Finance
R2	s138_eligible: enforceable debt + valid presentation + qualifying return code	Legal
R3	Statutory clock; alerts T-10, T-5, T-2 to named owner	Legal
R4	Segregation of duties — recommender ≠ approver	Architecture
R5	Block all action under insolvency moratorium	Finance / Legal
R6	No notice amount without lineage to source rows	Finance

Rules: versioned config (JSON), independently unit-testable, fail-closed (throw if prerequisite missing), changeable without prompt edits.

12. Limitation Clock Service



13. Security Architecture



14. Observability Architecture

Application Insights + OpenTelemetry (ARC.Agents/Middleware)

Span hierarchy:

- arc.cycle
 - arc.dealer
 - arc.executor.A1_Reconciliation
 - arc.tool.compute_net_exposure
 - arc.rules.evaluate { rule.set, rule.id }
 - arc.llm.explain { model, tokens }
 - arc.gate.depot_manager { suspended_utc, resumed_utc, wait_hours }

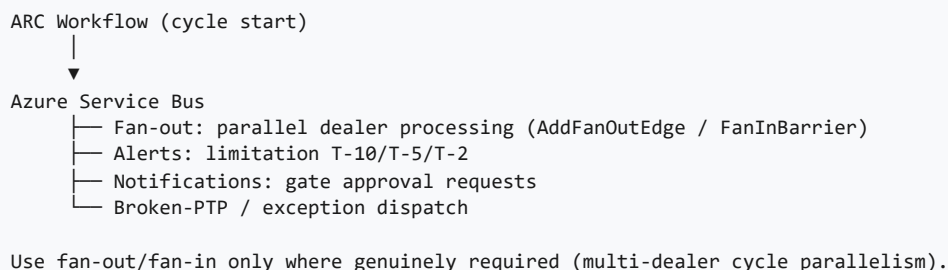
Custom dimensions (every span):

- cycle_id · dealer_urn · correlation_id · rule_version
- metric_contract_version · prompt_version

Immutable audit: GateDecision (SQL) + auditEvents (Cosmos)

Dashboard (pre go-live)	Purpose
Gate wait-time distribution	HITL SLA monitoring
Override rate by depot	System health (P2 regression detection)
R1b block rate by region	Reconciliation quality
Clock status histogram	Limitation risk
Self-verification failure rate	Draft quality

15. Messaging Architecture



16. Durable Workflow & Shadow Mode

Azure Functions
↓
Durable MAF Workflow (Microsoft.Agents.AI.Workflows.Durable)
↓
Cosmos DB Checkpoint (externalised – NOT in-memory only)
↓
Human Gate (RequestPort suspend, days/weeks)
↓
Process Restart → Different Instance → Resume from Checkpoint

RunMode: Shadow | Assisted | Live
IOutboundGate adapter: Shadow = compute everything, issue nothing (no notice despatch, no visit task push, no legal transaction). Enforced centrally – not scattered flags.

Idempotency: (cycle_id, dealer_urn, node) – retries cannot double-issue notice.

17. Recommended Solution / Project Structure

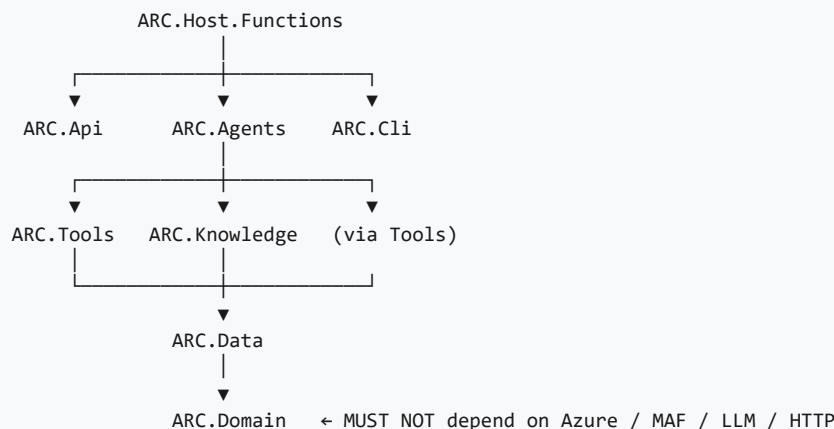
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ARC.sln
├── src/
│   ├── ARC.Domain/           ← Pure domain. NO Azure, MAF, LLM, HTTP.
│   │   ├── Entities/
│   │   ├── Metrics/         ← MetricContract, ExposureBreakdown, LineItemRef
│   │   ├── Rules/           ← IRule, R1a..R6, RuleConfiguration
│   │   ├── Clocks/          ← LimitationClock, ILimitationClockService
│   │   └── State/            ← RecoveryState
│   ├── ARC.Data/
│   │   ├── Sql/             ← EF Core / Dapper, read-only default
│   │   ├── Cosmos/          ← Checkpoints, cycleState, vector, auditEvents
│   │   └── Blob/             ← Documents, WORM SAS
│   ├── ARC.Knowledge/
│   │   ├── Graph/
│   │   ├── Retrieval/
│   │   └── Extraction/       ← Document Intelligence wrappers
│   ├── ARC.Tools/
│   │   ├── ToolRegistry.cs
│   │   └── Mcp/
│   ├── ARC.Agents/
│   │   ├── Executors/       ← A1..A8
│   │   ├── Workflows/       ← OdosCycleWorkflow, Section138Workflow
│   │   └── Middleware/      ← PII, correlation, telemetry
│   ├── ARC.Host.Functions/   ← Durable workflow host
│   ├── ARC.Api/              ← Gates, dashboards, NLQ endpoints
│   └── ARC.Cli/              ← Scenario runner S1..S9
├── tests/
│   ├── ARC.Domain.Tests/     ← Rules + clock (highest value)
│   ├── ARC.Agents.Tests/     ← Workflow topology, checkpoint resume
│   └── ARC.Eval/             ← Golden-set harness (~200 cases)
```

18. Responsibility of Each Project

Project	Responsibility
ARC.Domain	Entities, metric contracts, rules (R1–R6), limitation clock, RecoveryState, enums, interfaces. Zero infrastructure dependencies.
ARC.Data	Azure SQL repositories, Cosmos checkpoint/store, Blob document repository, GateDecision persistence.

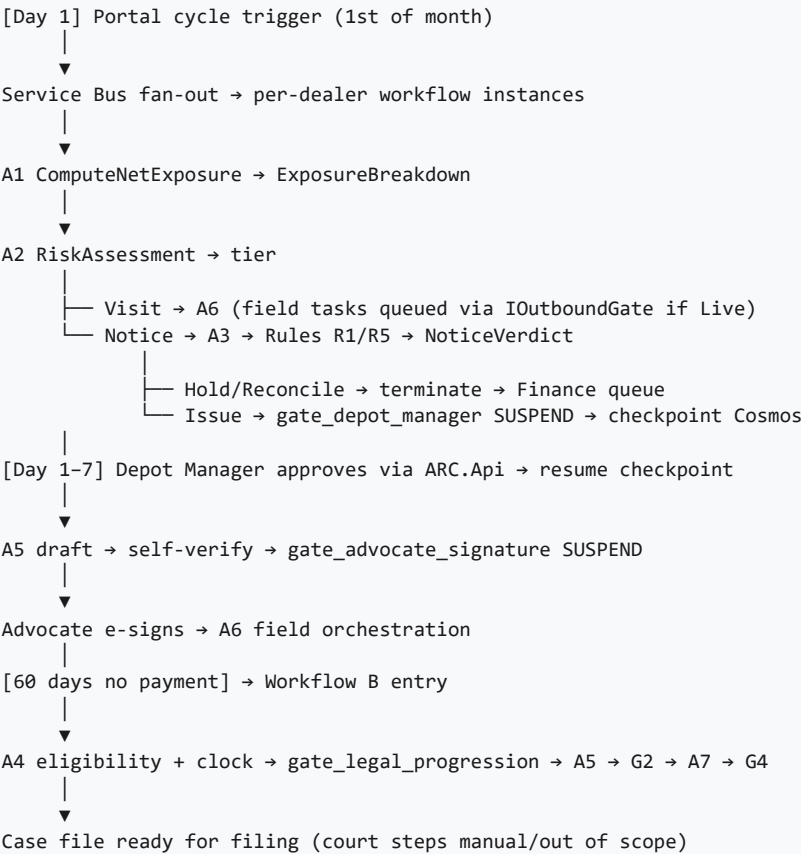
ARC.Knowledge	Knowledge graph traversal, hybrid retrieval (graph + keyword + vector), Document Intelligence extraction wrappers.
ARC.Tools	Governed AI tool surface: ComputeNetExposure, CheckSection138Eligibility, GetLimitationClock, SearchDocuments, TraverseGraph. MCP server.
ARC.Agents	MAF executors A1–A8, Workflow A/B graphs, RequestPort gates, middleware (PII, OTel), IOutboundGate.
ARC.Host.Functions	Azure Functions durable host binding, cycle triggers, Service Bus consumers, checkpoint resume.
ARC.Api	External API — human gate responses, dashboards, NLQ (A8), health. Container Apps.
ARC.Cli	Local scenario execution S1–S9, shadow mode runs, demo harness.
ARC.Domain.Tests	Rules R1–R6, limitation clock edge cases, metric contract lineage.
ARC.Agents.Tests	Workflow wiring, conditional edges, gate suspend/resume, idempotency.
ARC.Eval	Golden-set evaluation (~200 labelled cases), acceptance metric targets.

19. Dependency Flow



Dependency rule: ARC.Domain references nothing external.
If a rule or clock needs HttpClient, the design is wrong.

20. End-to-End Request / Workflow Flow



21. Architecture Decisions

Decision	Rationale (Source)
MAF WorkflowBuilder graphs, not chat loops	Multi-day durable process with explicit gates
Eight bounded executors (A1–A8), not one giant agent	Deterministic/probabilistic separation per sub-problem SP1–SP9

Rules in ARC.Domain as versioned code, not prompts	R1–R6 must be independently testable; Legal/Finance owned
LimitationClockService separate from A4 LLM	Zero statutory misses; dates never from model
Cosmos DB external checkpoint store	15+ day cycles; resume on different instance
No SubmitNotice tool	Notices emitted by workflow post-gate, not model tool call
IOutboundGate for Shadow/Live	Central enforcement — compute everything, issue nothing in shadow
ARC.Domain has zero infra deps	Testability; rules/clock pure logic
Read-only SQL default	Agents query; business services write
MCP + AIFunction dual exposure	Same governed contracts for agents and external clients
Service Bus fan-out for dealer parallelism	~2,500 dealers per cycle; parallel scoring/extraction
Fail-closed everywhere	Missing prerequisite → block, never default Pass

22. Items Explicitly Required by Source Document

- Two MAF workflows (A: ODOS, B: Section 138) sharing A1 upstream
- Four RequestPort HITL gates with Cosmos checkpointing; expiry ≠ approval
- Eight executors A1–A8 with decomposition as specified
- Rules R1–R6 versioned, fail-closed, unit-testable
- LimitationClockService with T-10/T-5/T-2 alerts to named owner
- Five governed tools (no SubmitNotice)
- Metric contract net_recoverable_exposure with R6 LineltemRef lineage
- Document Intelligence with confidence thresholds (0.90/0.90/0.85)
- Hybrid retrieval: graph + keyword + vector (Cosmos diskANN)
- Shadow / Assisted / Live run modes via IOutboundGate
- Idempotency key (cycle_id, dealer_urn, node)
- GateDecision immutable audit + Cosmos auditEvents (8-year)
- PII redaction middleware before model calls
- RBAC server-side — TSI cannot read other regions
- OpenTelemetry spans with cycle_id, dealer_urn, correlation_id, rule_version, prompt_version
- Demo scenarios S1–S9 and golden-set evaluation targets
- Solution structure ARC.sln with 8 src + 3 test projects as specified
- WORM Blob 8-year retention for legal artefacts
- SP1–SP3 upstream sequencing; A1 shadow before live notice influence

23. To Be Confirmed (Source Does Not Fully Define)

Item	Notes
Exact statutory periods (NI Act windows)	Confirm with Legal counsel; hold as versioned config (illustrative 30/15/30 in source)
Clock anchor: memo received vs issue date	Configured anchor — must be explicit
Second bounce — which dishonour anchors clock	Policy decision required
Notice-by on Sunday/gazetted holiday	Behaviour must be documented
R1b denominator (claim_amount vs gross_open_AR)	Reconcile rule text with implementation
R1c PTP grace period duration	Not numerically specified
A2 scoring model / tier thresholds	Tier enum defined; boundaries TBC
ASR hard-floor discard threshold	S7 uses 0.72 for confirmation tier only
15th timeline freeze mapping to agent workflow	In current Process B; not in MAF topology
Arrest warrant / court proceeding automation	Portal column exists; no agent beyond filing readiness
Physical courier chain automation	Manual steps remain; system blocks ineligible pre-courier
Hold vs Reconcile termination semantics	Both terminate without G1; clarify routing difference
MAF API namespace stability	Verify against current MAF 1.0 GA docs

— End of Document —

ARC Solution Architecture Document v1.0 | 28-Aug-2026

No implementation code — architecture only